DIGITAL LOGIC & DESIGN LAB

RA: Maham Tabassum

INSTRUCTORS

- **Instructor: Ms. Areeba Rajput**

  Office Hours: Tues – 1 - 2 PM

- **Lab RA: Ms. Maham Tabassum**

  Office Hours: MW – 10:30 – 11:30 AM

ATTENDANCE POLICY

- Attendance will be marked **within the first 5 minutes**.
- **Keep check on your attendance** on PSCS.
- Instructor(s) will not be able to make any changes **after 1 week**.

LAB MANUAL SUBMISSION

a. For each week's lab, manual/worksheet (depending on the lab) will be **submitted in softcopy before the following Lab**.

b. **Late submission policy**: Deadline for submission of each lab is 1 week.

c. Late submissions will be penalized by a deduction of **5% per day**. Not accepted after 2 weeks of the lab execution.

PLAGIARISM

Case will be reported to the concerned department. Both copier and copiee will be charged guilty.

SYLLABUS

Syllabus is available on LMS Canvas & Simple Syllabus:

Course Syllabus - Simple Syllabus

LAB GUIDELINES

- Students are **not allowed to displace/relocate any of the equipment** of the lab without RA's permission.
- **No Food and Drinks** allowed.
- Keep your **bags outside in the rack**.
- **Bring your own *plug/multiplug*** for your own laptops – no extension/plug will be issued.

LAB GUIDELINES

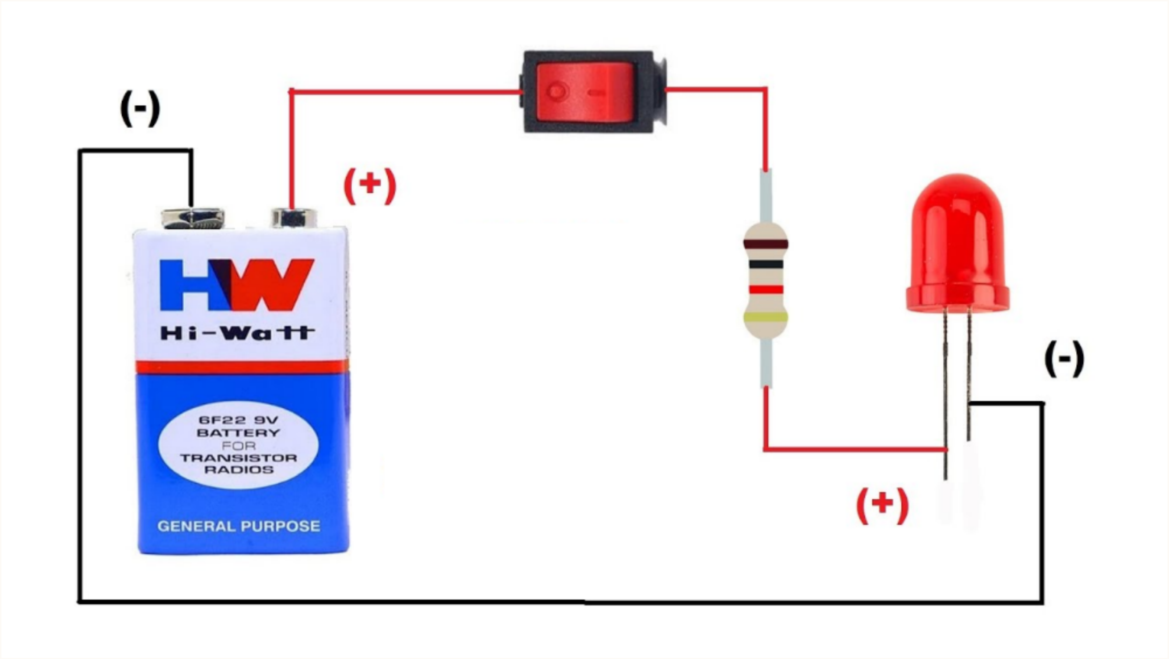
- Once you complete your **in-lab tasks**, please get them checked by the RA during lab hours to ensure they are graded.
- Your performance on in-lab tasks will be recorded on your ***Rubric Sheet***.
- Rubric sheets follow a level-based marking system, depending on the quality of performance observed in the lab.
- By the end of the lab:

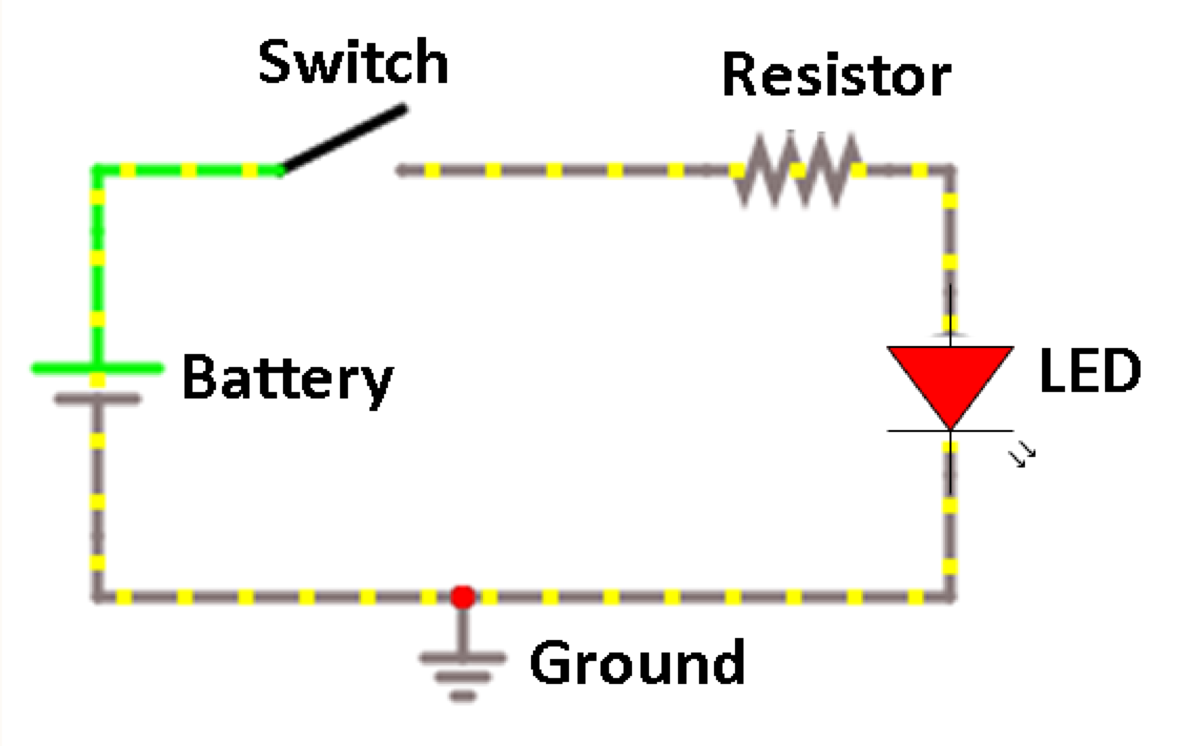
  - Make sure your **workstation is clean**.
  - All **electrical components used must be sorted and placed back** in the relevant trays (if not faulty).
  - Any **faulty components** must be placed in the *"Consumables"* drawer.
- Before leaving, please get a ***clearance sign*** on your Rubric Sheet; this is required for your lab to be graded.

LAB 01

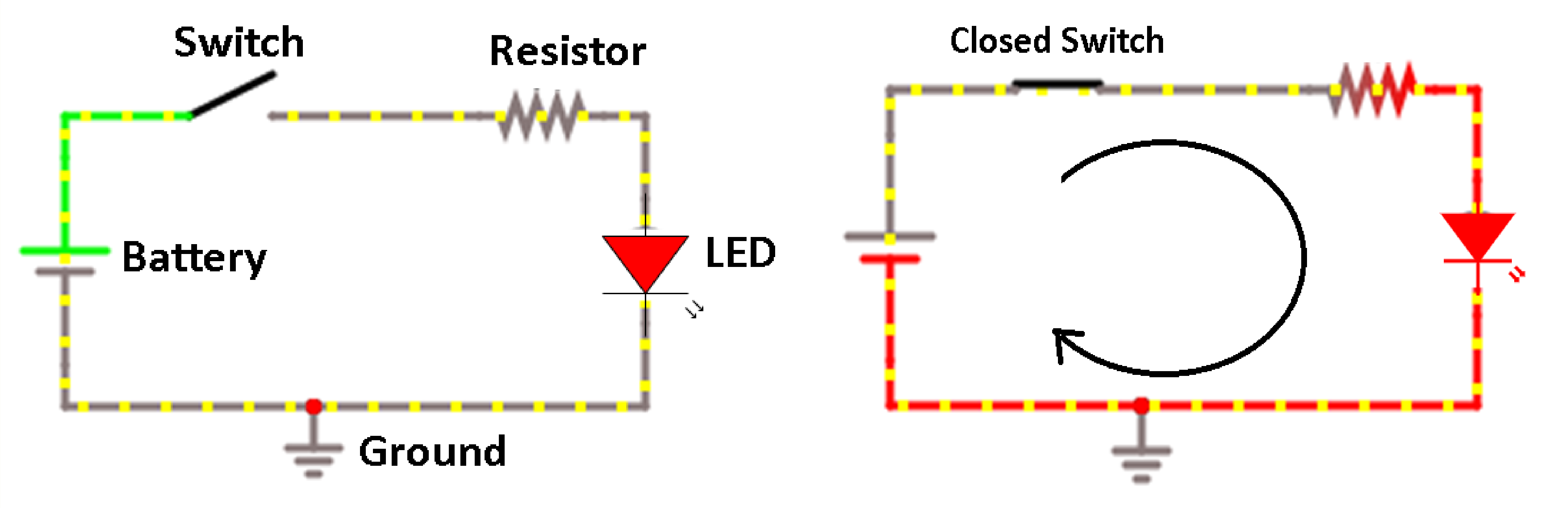
INTRODUCTION TO CIRCUITS & LAB EQUIPMENT

ELECTRIC CIRCUIT & COMPONENTS





CLOSED & OPEN SWITCH, DIRECTION OF CURRENT



POWER SUPPLY

Powers up the circuit

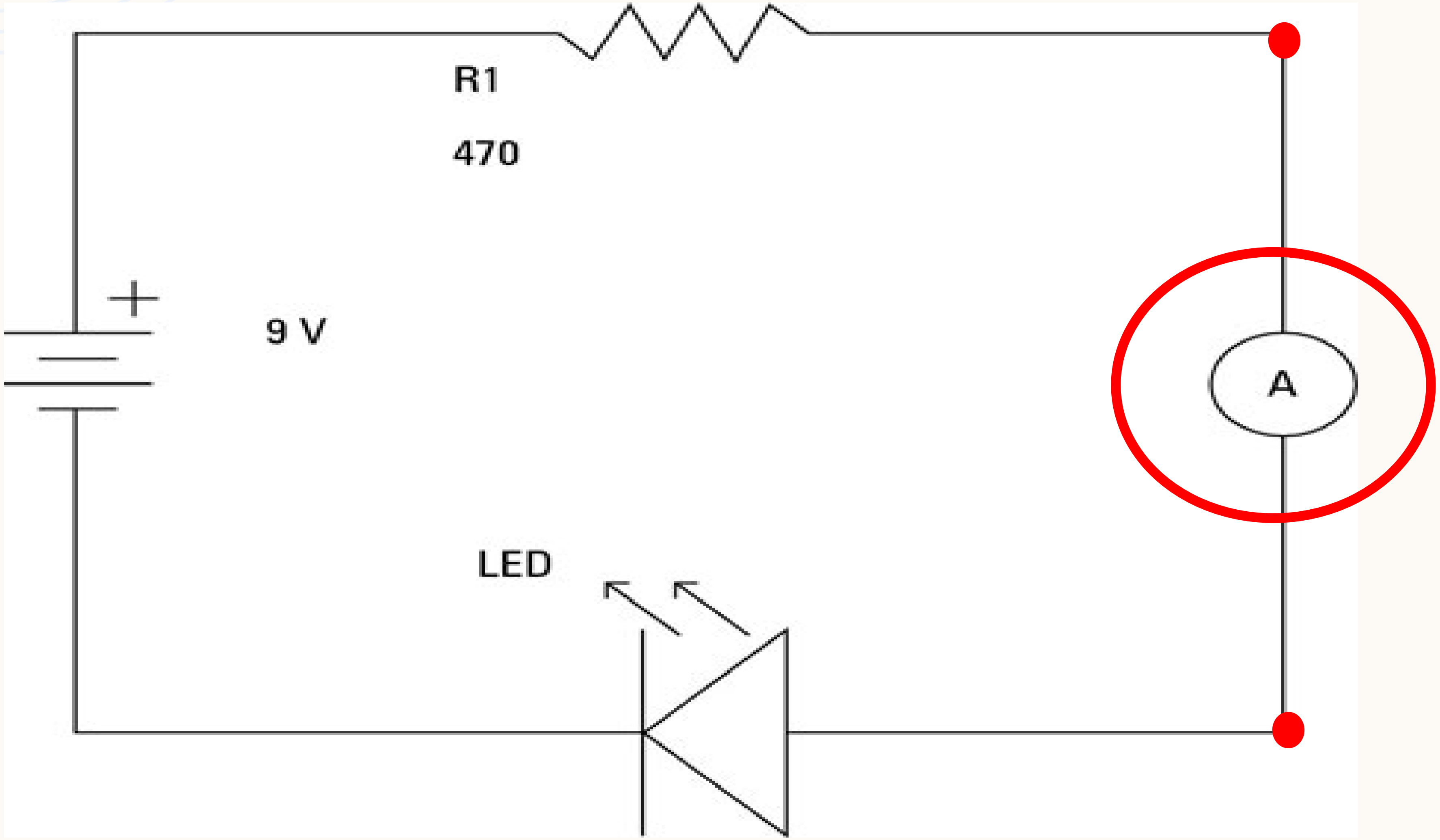
DIGITAL MULTIMETER

Measures Current and Voltage

HOW DO WE MEASURE ELECTRIC CURRENT IN A CIRCUIT?

💡 *Which instrument do we use to measure current?*
💡 *How should it be connected in a circuit – series or parallel?*
💡 *Any safety measures of the lab instrument while measuring electric current?*

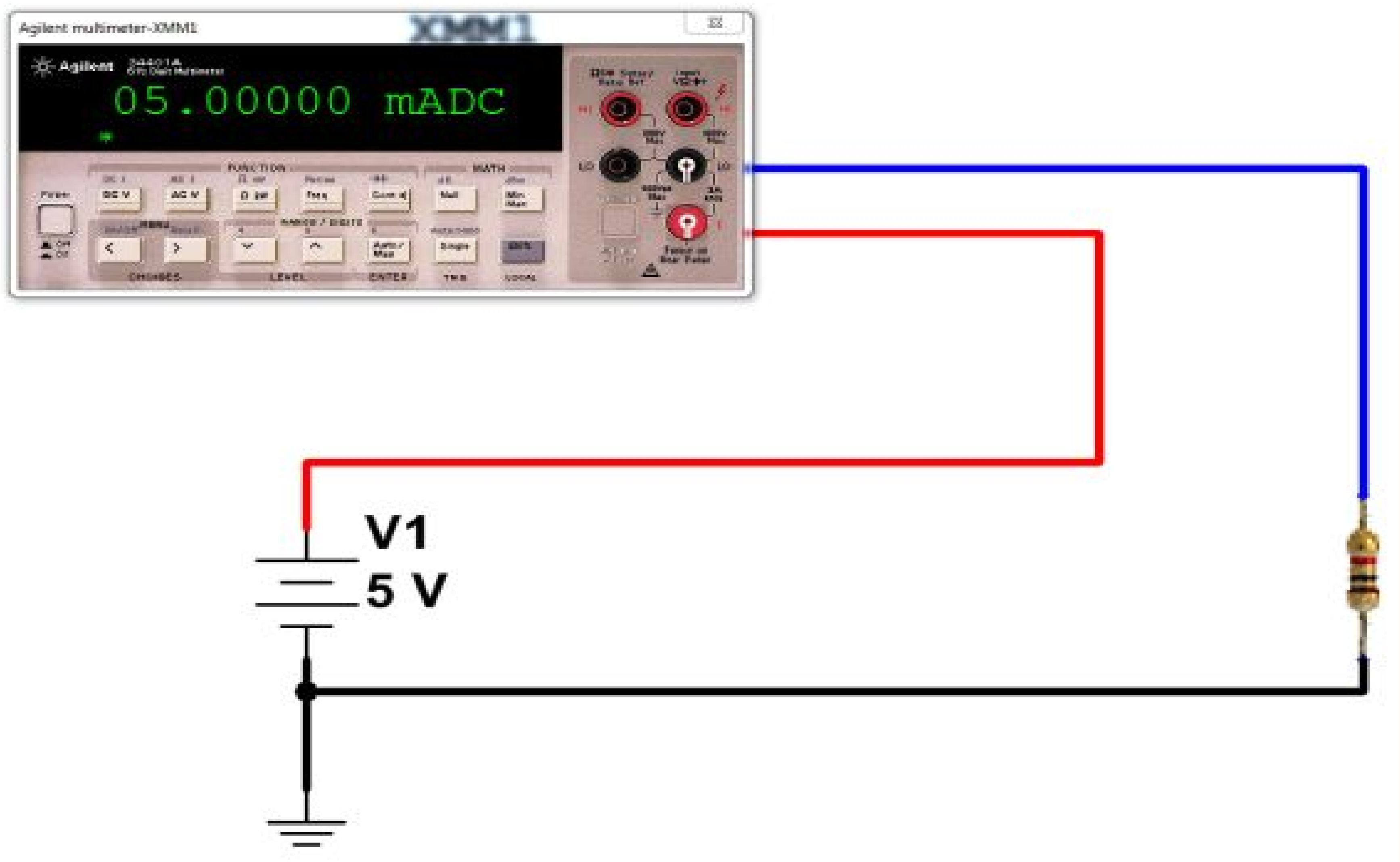
MEASURING ELECTRIC CURRENT IN A CIRCUIT



- **Open** the circuit at one point.

- Connect the ammeter across the gap—one probe to each open end—so the **current passes through the meter** (series).

MEASURING ELECTRIC CURRENT USING DMM



- Place DMM **in series** with the circuit.
- Set DMM to **Current (A) mode**.

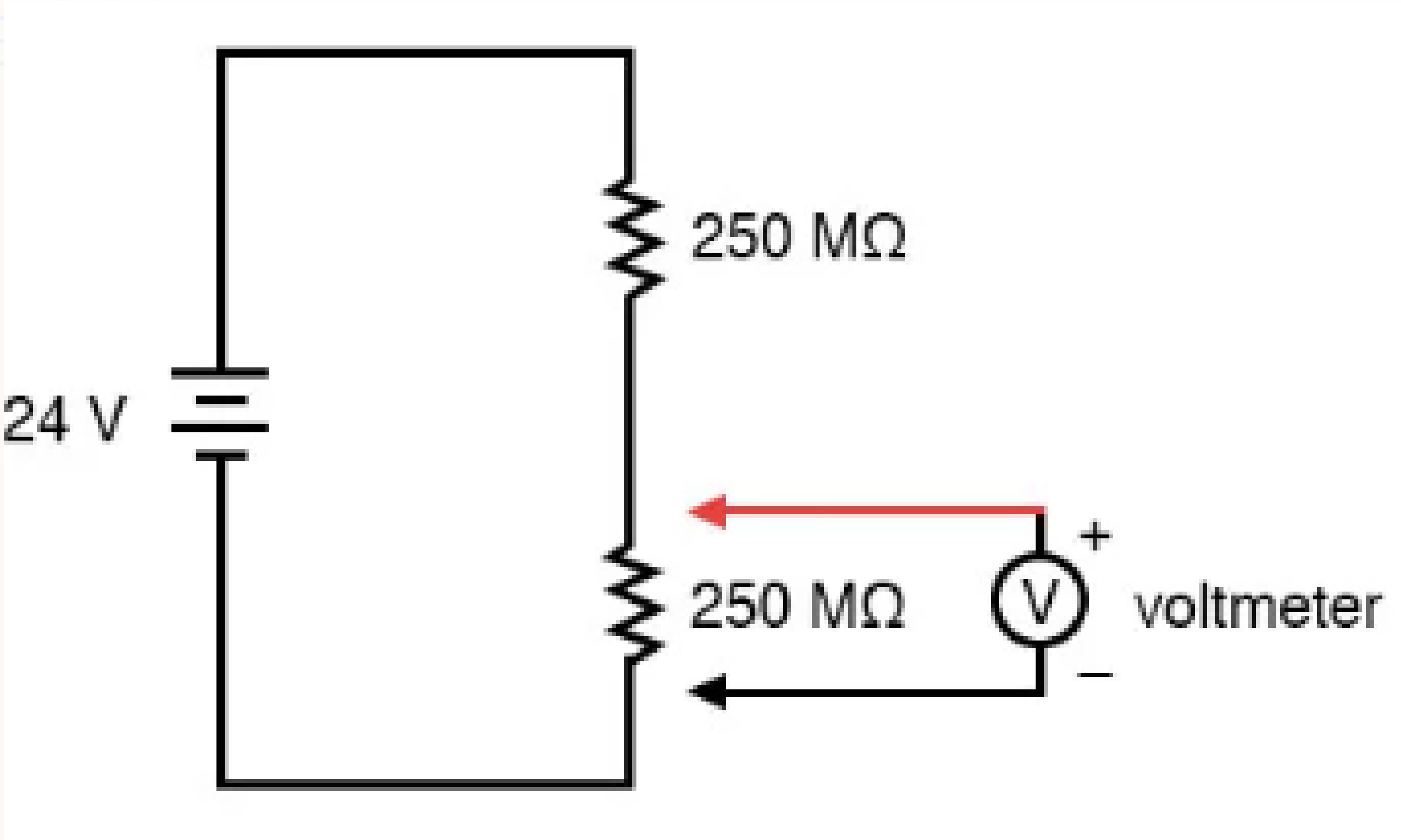
⚠️ **Insert probes in correct ports** (COM and A/mA).

⚠️ **Check fuse** – high current can blow it.

HOW DO WE MEASURE VOLTAGE IN A CIRCUIT?

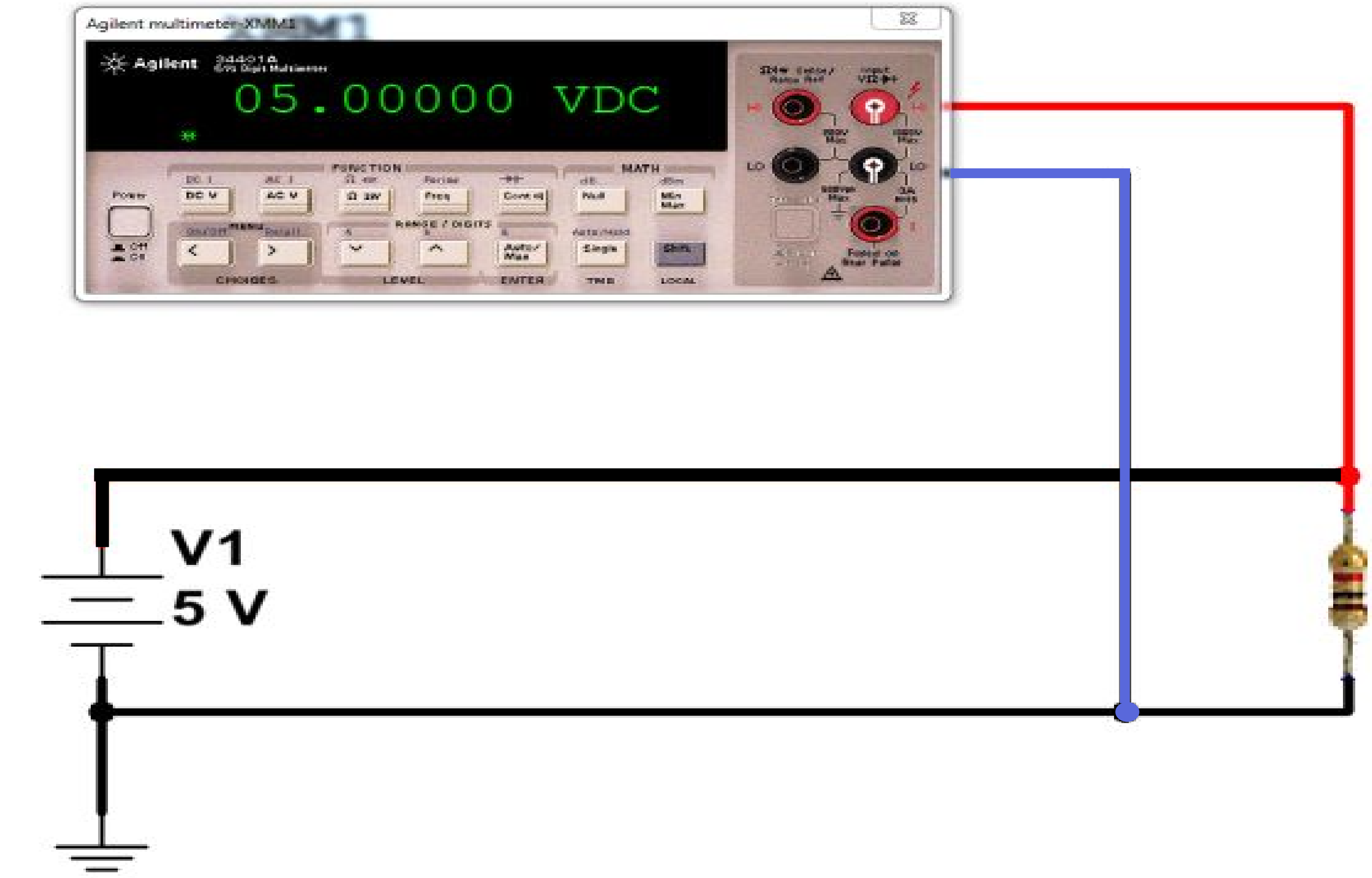
💡 *Which instrument do we use to measure voltage?*
💡 *How should it be connected in a circuit – series or parallel?*
💡 *Any safety measures of the lab instrument while measuring voltage?*

MEASURING VOLTAGE IN A CIRCUIT



- **Do not break** the circuit.
- Connect the voltmeter **across the component** – one probe on each side – so it measures the potential difference (parallel).

MEASURING VOLTAGE USING DMM



- Place the instrument **in parallel** across the component.
- DMM to **Voltage (V) mode**.

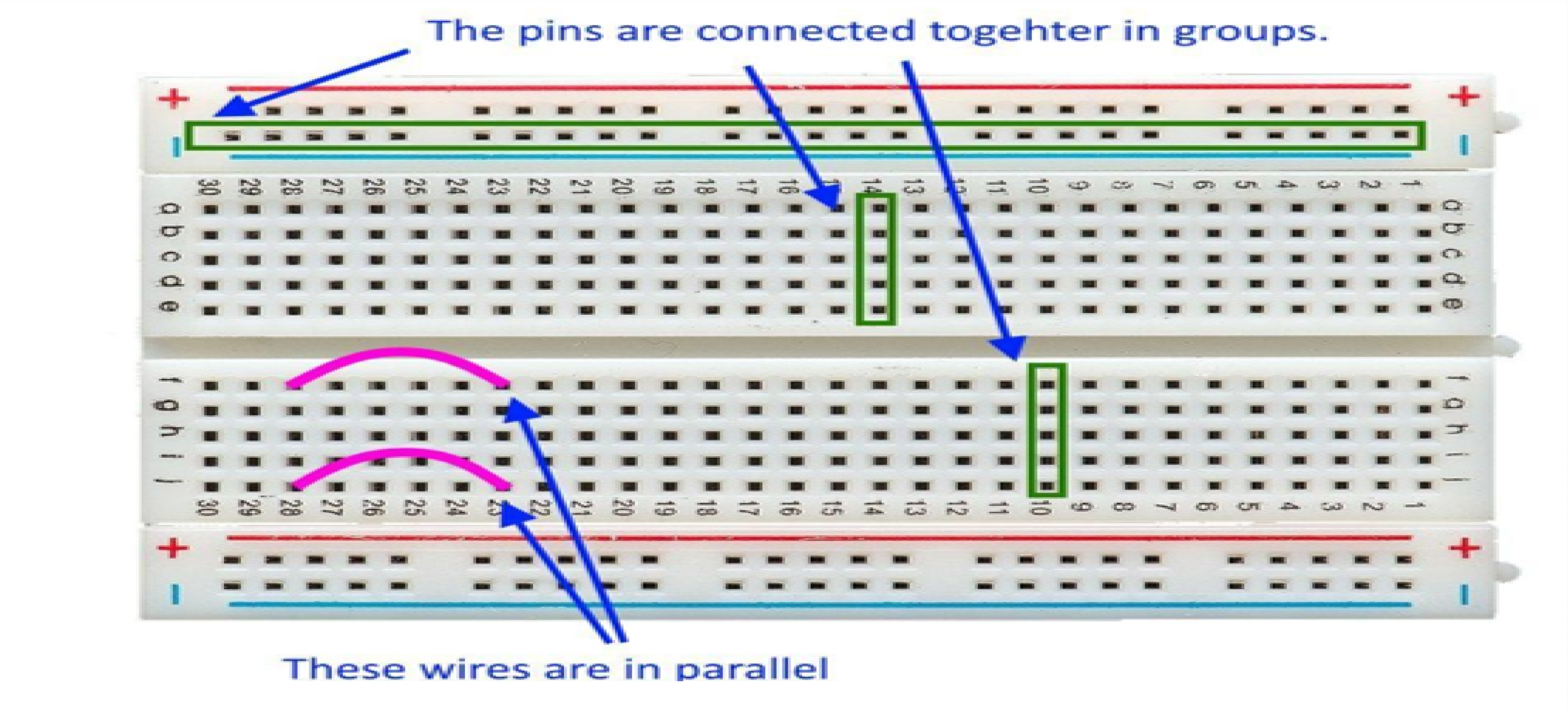
⚠️ Choose **correct DC/AC setting**.

⚠️ **Never use Current mode** when measuring voltage.

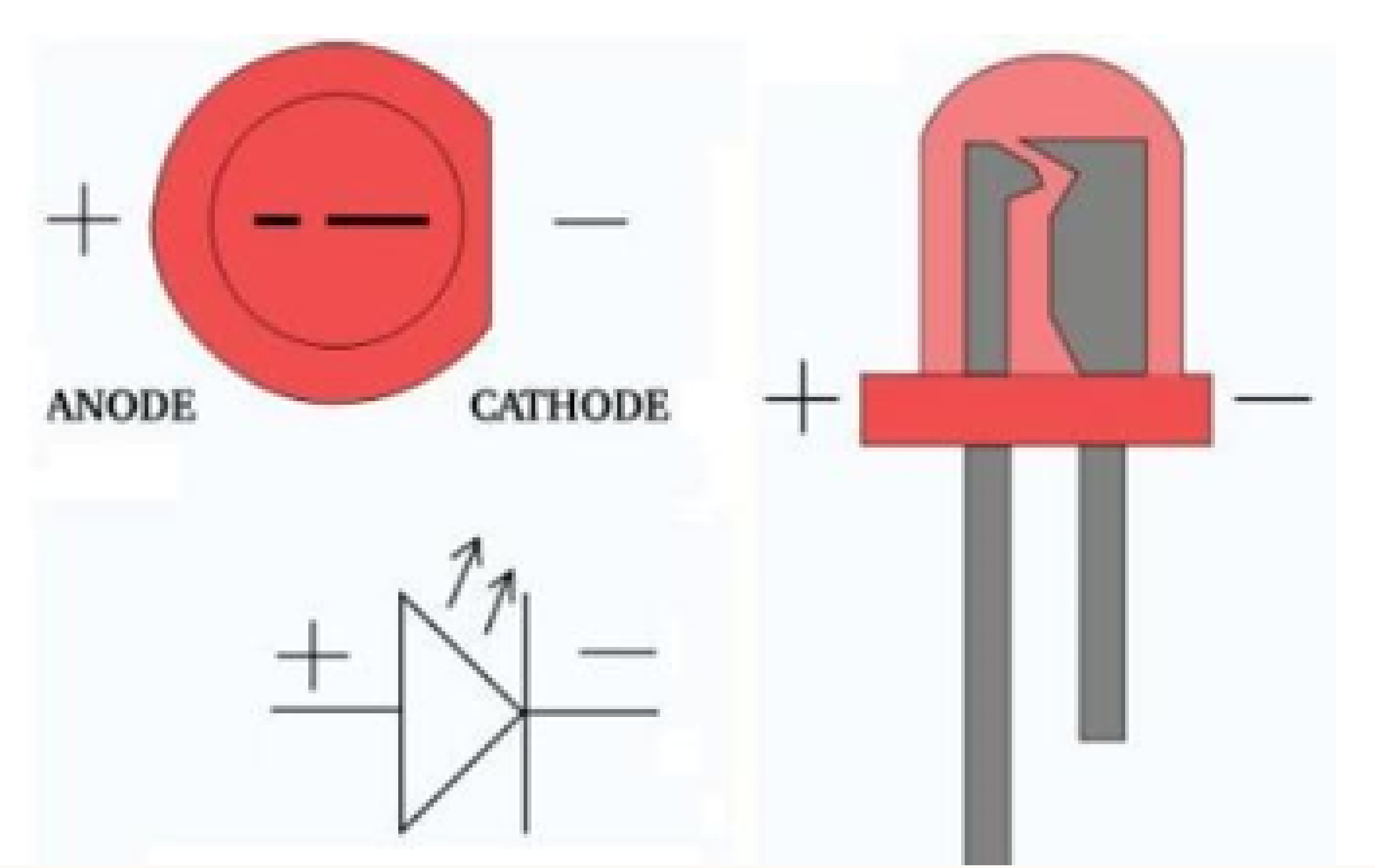
VOLTAGE IN DIGITAL LOGIC

ON	OFF
‘1’	‘0’
HIGH	LOW
5V (typically)	0V

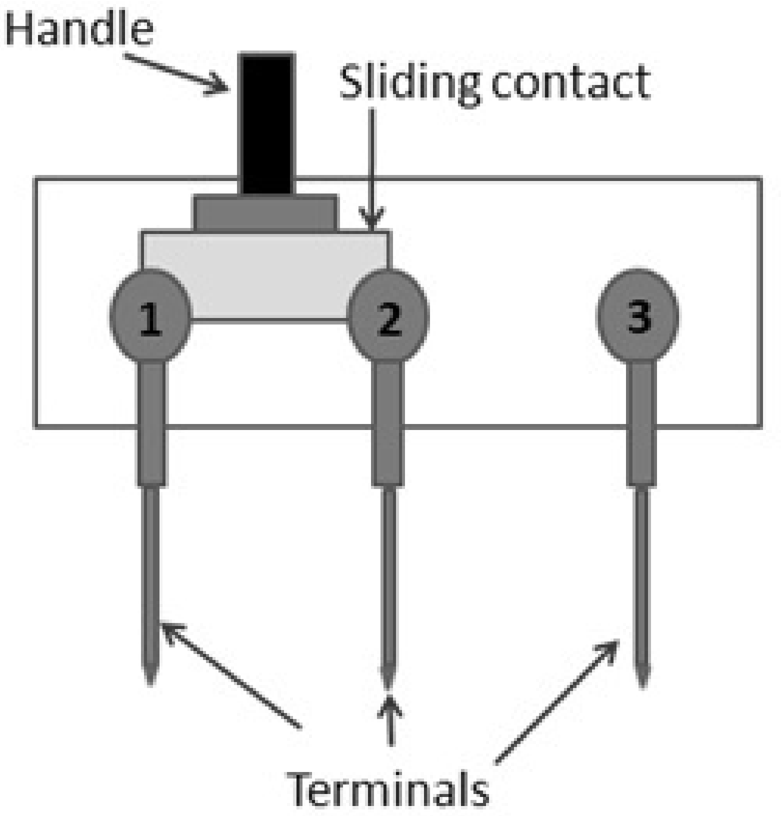
BREAD BOARD


The pins are connected togehter in groups.
These wires are in parallel

LIGHT EMITTING DIODE (LED)

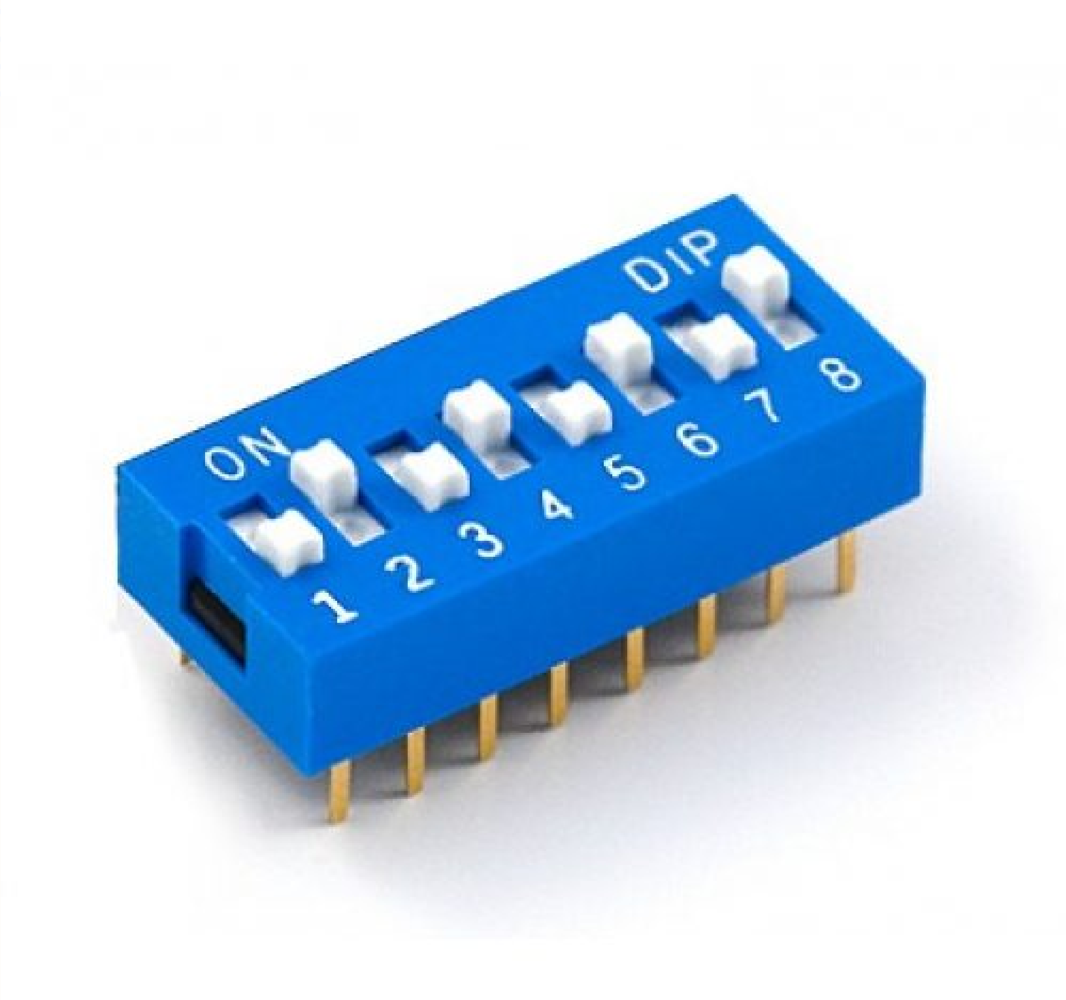


SLIDE SWITCH



- The center pin is the **common pin** whereas the extreme pins are the **toggling state 1** and **toggling state 2**.
- Connect common pin to the circuit.
- Connect one of the toggling pins to the positive end of the power source, and leave the second toggling pin

DIP SWITCH



- Special type of switch: consists of **many single contact switches**.
- Shape is similar to IC – **connected in middle two lines (e & f)** of the breadboard.

CLEARANCE

- Clean up your workstation.
- Turn off all the lab equipment.
- Put all the components back to the relevant labelled tray *if not faulty*.
- Faulty components must be put in '*Consumables*' box.
- Get your **CLEARANCE SIGN** by the RA.
- Put your breadboards in the designated drawer.